EXPERIMENT 10: BLOCKS WORLD PROGRAM

AIM

To implement a Blocks World program that represents block stacks, sets a goal configuration, and performs block moves.

PROCEDURE

1. Initialize the blocks on separate stacks.

2. Display the initial state and set the goal configuration.

3. Move blocks according to the program and display the state after each move.

CODE

Python Program

```
class BlocksWorld:
    def __init__(self, num_blocks):
        self.state = [[block] for block in range(num_blocks)]
        self.num_blocks = num_blocks

    def display_state(self):
        for stack in self.state:
            print(f"Block(s) on stack: {stack}")

    def find_block(self, block):
        for stack in self.state:
            if block in stack:
                return stack
        return None

    def move(self, block, destination):
        source_stack = self.find_block(block)
        destination_stack = self.find_block(destination)
        if source_stack is None or destination_stack is None:
            print(f"Invalid block {block} or destination {destination}.")
            return
        source_stack.remove(block)
        destination_stack.append(block)
        self.display_state()

    def goal_state(self, goal):
        self.state = goal
        print("Goal state set.")
        self.display_state()

def main():
    blocks_world = BlocksWorld(3)
    print("Initial state:")
    blocks_world.display_state()

    goal = [[0, 1], [2]]
    blocks_world.goal_state(goal)

    print("\nPerforming Moves:")
    blocks_world.move(0, 2)
    blocks_world.move(1, 2)
    blocks_world.move(2, 0)

if __name__ == "__main__":
    main()
```

OUTPUT

```
Initial state:
Block(s) on stack: [0]
Block(s) on stack: [1]
Block(s) on stack: [2]
Goal state set.
Block(s) on stack: [0, 1]
Block(s) on stack: [2]
Performing Moves:
Block(s) on stack: [1]
Block(s) on stack: [2, 0]
Block(s) on stack: []
Block(s) on stack: [1]
Block(s) on stack: [2, 0]
Block(s) on stack: []
Block(s) on stack: [1, 2]
Block(s) on stack: [0]
Block(s) on stack: []
```

RESULT

Thus, the Blocks World program was executed to represent stacks, set a goal state, and display the resulting block configurations.